LM in Economics and Econometrics

[2025–2026]

Consumer Theory

Lecture notes

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Preface

These notes accompany the consumer-theory block of the course. They use the same core package as the slide theme and the problem-set template, so a definition looks identical wherever it appears.

1 Preferences

1. A preference relation $\succsim$ on a choice set X is a binary relation read as "at least as good as". *Gianluca: add the axioms table here*

2. Two derived relations follow. Strict preference $\succ$ holds when $x \succsim y$ but not $y \succsim x$; indifference $\sim$ holds when both directions hold.

Definition 1.1 (Rational preferences). A preference relation $\succsim$ is rational if it is complete and transitive.

Proposition 1.2. If $\succsim$ is rational then $\succ$ is irreflexive and transitive, and $\sim$ is an equivalence relation.

Proof. Immediate from completeness and transitivity. □

> **Box 1 — Why continuity matters**
>
> Rationality alone does not deliver a utility representation. Continuity rules out the lexicographic counterexample, and with it the jump that breaks any candidate representation.

2 Utility representation

3. A function $u : X \to \mathbb{R}$ represents $\succsim$ when $x \succsim y \iff u(x) \geq u(y)$.

Theorem 2.1 (Debreu). A rational and continuous preference relation on a connected subset of $\mathbb{R}^n_+$ admits a continuous utility representation.

$$\boxed{x \succsim y \iff u(x) \geq u(y)}$$

Remark. The representation is unique only up to a strictly increasing transformation.

> **Box 2 — Ordinal, not cardinal**
>
> Nothing in the representation licenses comparing utility differences across people, or even across bundles for one person.

3 Demand

4. The Walrasian demand correspondence solves $\max_x u(x)$ subject to $p \cdot x \leq w$. **[NB: add the Lagrangian]**

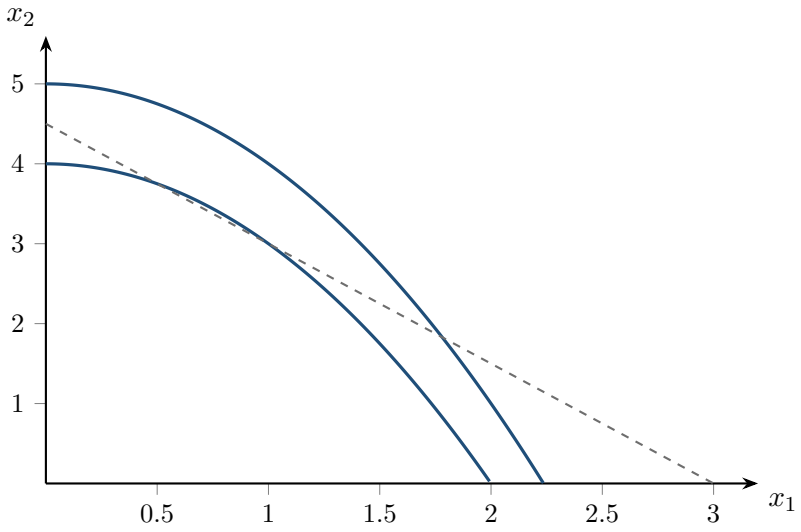


Figure 1: Indifference curves and the budget line.

$x^*(p, w) = \dfrac{w}{2p}$ under Cobb–Douglas with $\alpha = 1/2$.

A Notation

A short reference for the symbols used above.

- $\succsim$ — weak preference (at least as good as)
- $\succ$ — strict preference
- $\sim$ — indifference
- UCS — upper contour set